Trainings

General Information

Industrial Applications

The Cummins® QSK45 and QSK60 engines require a minimum cooling system pressure cap rating of 76 kPa [11 psi]. This provides a positive coolant pressure at the water pump inlet.

The QSK45 and QSK60 industrial engines use separate main engine water pumps but use the same low temperature aftercooling (LTA) water pump in the two-pump, two-loop cooling system with low temperature aftercooling. This means that the engine uses two cooling loops consisting of a main engine loop and an LTA loop. The main engine loop controls the jacket water circuit. It runs at higher temperatures and is sometimes referred to as the high temperature circuit. The low temperature aftercooling loop runs at lower temperatures and is sometimes referred to as the low temperature circuit.

The QSK45 and QSK60 industrial engine cooling circuits use a common surge tank. The tank is designed to reduce turbulence within the tank through the use of baffles. During engine operation, small quantities of coolant from both circuits will mix in the surge tank.

The QSK45 engine often uses the QSK60 main engine water pump in cases where additional output is required, but the QSK45 main engine primary water pump can **not** be used on the QSK60 engines.

The water pumps are centrifugal pumps with cast iron impellers. The main water pump is gear driven by a spline coupling that connects it to the water pump drive. The LTA water pump is gear driven by a spline coupling that connects it to the hydraulic pump drive.

Each water pump contains two ball-type bearings. The bearings are lubricated with pressurized engine oil. The QSK45 and QSK60 primary water pumps contain a front, large ball bearing that is a single-row bearing.

The impeller and the water inlet connection have a machined angle that is critical to the performance of the pump. The impeller and water inlet connection are **always** machined to the same angle.

The water pump shaft contains **only** one retaining ring and requires a spacer between the two bearings.

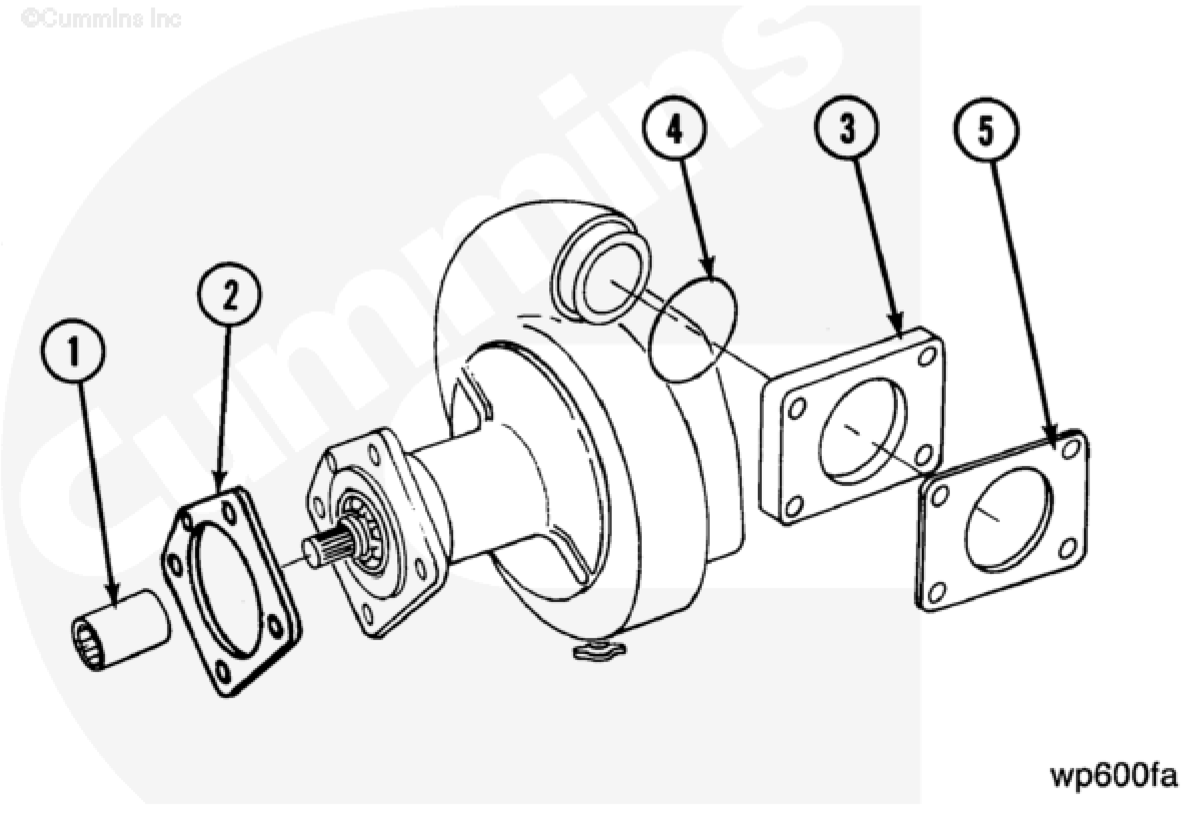
The water pump contains an oil seal and a water seal. The cavity between these seals has a vent. The vent prevents contamination of the lubricant or coolant if a seal leaks. Oil and coolant seeping from the vent will **not** harm the operation of the pump. Check the vent for any obstruction at each scheduled maintenance interval. The recommended interval is 250 hours or 6 months of operation.

A special tool **must** be used to install the one-piece water seal so that the seat is installed at the correct dimension from the water pump body. Correct installation results in proper spring tension.

Two vent lines from the engine to the surge tank are used on the QSK45 and QSK60 single-stage industrial engines. One line vents the aftercooler circuit and one line vents the jacket water circuit. Vent lines from the left-front and right-front aftercoolers are connected to a T-fitting at the thermostat housing on the left-bank side. **Only** the front aftercoolers have vent lines. The rear aftercoolers each have a petcock that needs to be kept open during filling. The jacket water vent is located at the thermostat housing on the right side.

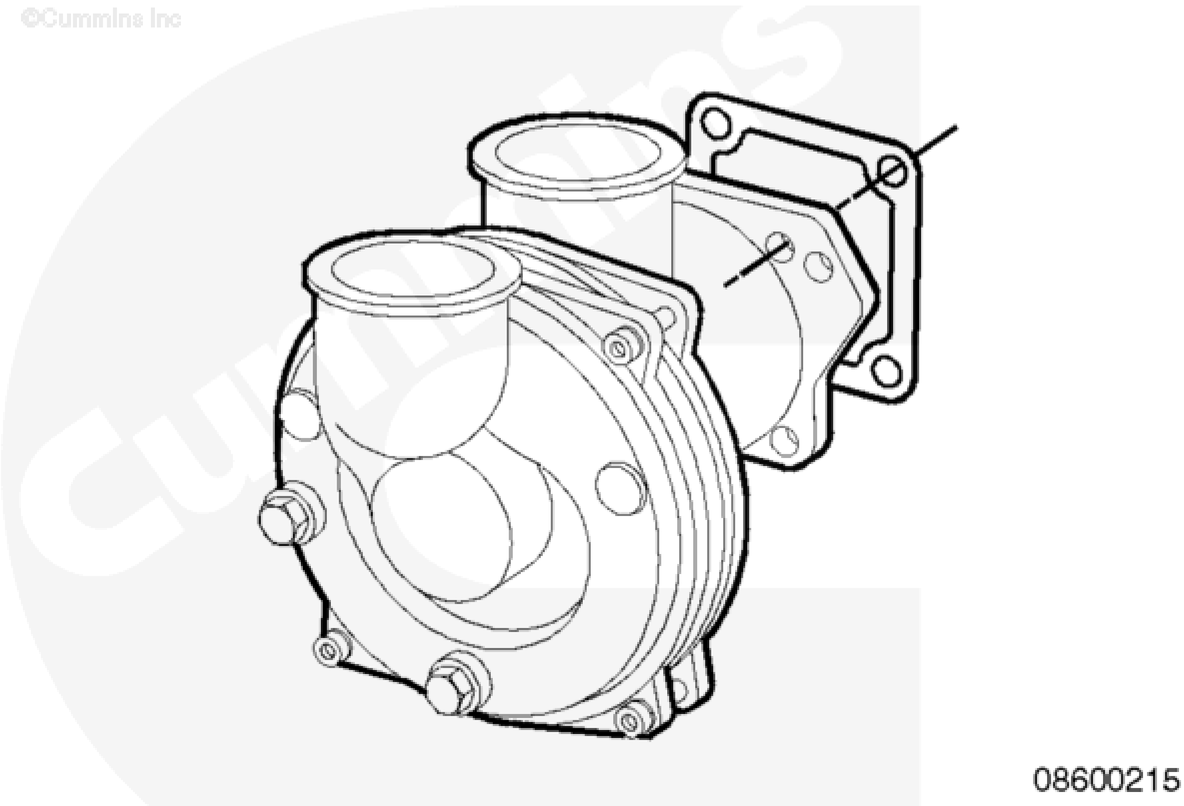
Two additional vent lines from the engine to the surge tank are required for the QSK60 two-stage industrial engine. One additional vent line goes from each of the two intercoolers on the engine to the surge tank.

The QSK45 and QSK60 industrial engines use separate main engine water pumps but use the same LTA water pump in the two-pump, two-loop cooling system with low temperature aftercooling.



Main Engine Water Pump (Two Pump/Two Loop)

The main engine cooling loop's primary purpose is cooling the engine block. Each pump sends coolant to an independent radiator.



Low Temperature Aftercooling Pump (Two Pump/Two Loop)

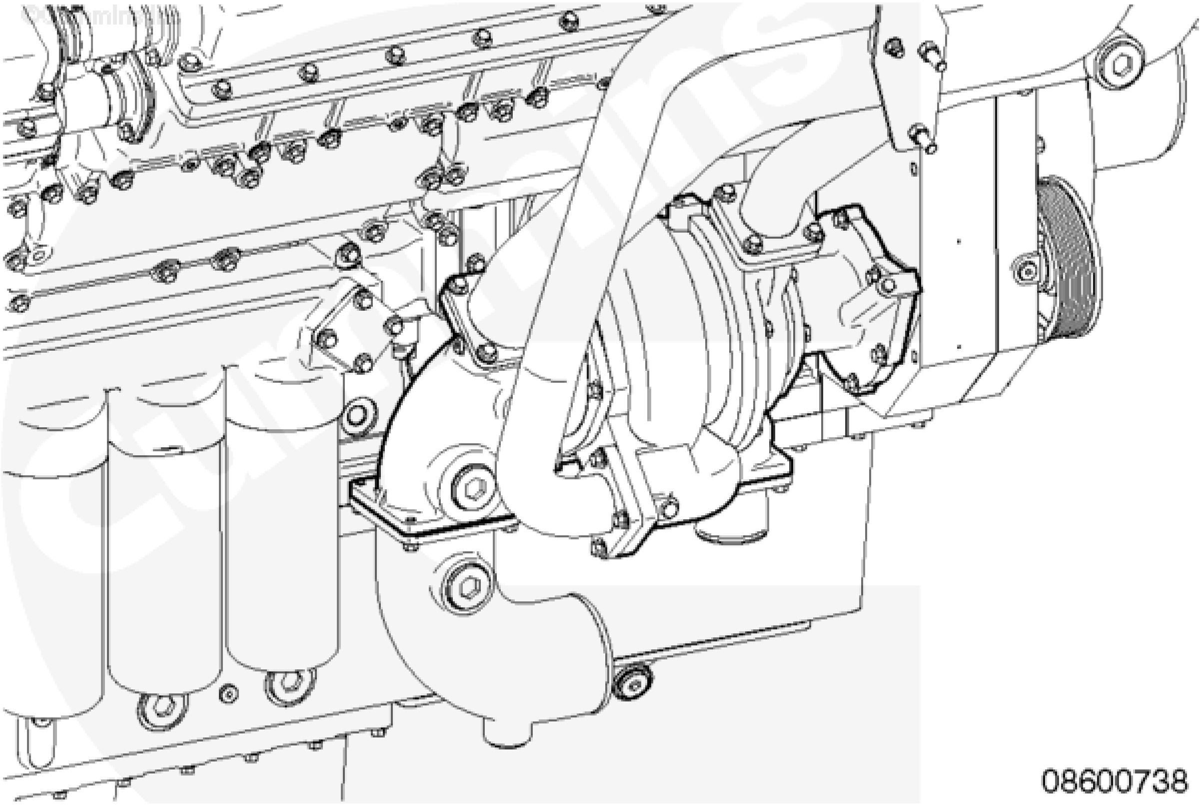
The LTA cooling loop's primary purpose is cooling the intake air. Each pump sends coolant to an independent radiator.

The LTA radiator contents are normally cooler than that of the engine radiator. The LTA radiator should be positioned so that it receives the cool air before the engine radiator.

with Electronically Actuated Injector

Engines with electronically actuated injectors are equipped with a thermostat housing that allows thermostat replacement without the removal of the entire housing. This thermostat housing has cover plates that can be removed allowing access to the thermostats.

QSK60 engines use separate main engine and LTA water pumps that are on a common shaft. This means that the engine uses two different cooling loops, but has **only** one water pump assembly on the engine. The pump contains two different impellers, one for each cooling loop. The low temperature aftercooler impeller is the one more forward on the engine. With the exception of the water pump and thermostat housing, the cooling system for engines with electronically actuated injectors functions like the cooling system on an engine with mechanically actuated injectors.



Water Pump (Two Pump/Two Loop with Electronically Actuated Injectors)

Marine Applications

The engine cooling system is a two-pump, two-loop design similar to the industrial version. There are three available cooling options: heat exchanger, keel-cooled, and radiator.

The heat exchanger uses a titanium plate design that is mounted on the front of the engine. The jacket water and LTA circuits are both located in the same heat exchanger and divided by a pair of blanking plates. The sea water pump is mounted off the front of the hydraulic gear drive. The LTA pump is mounted on the rear of the hydraulic gear drive. The LTA pump is different between the heat exchanger and keel-cooled engines. There is a unique LTA pump for the heat exchanger. The keel-cooled pump is the same as it is for industrial engines. A service tool is integrated in the heat exchanger assembly to assist with servicing.

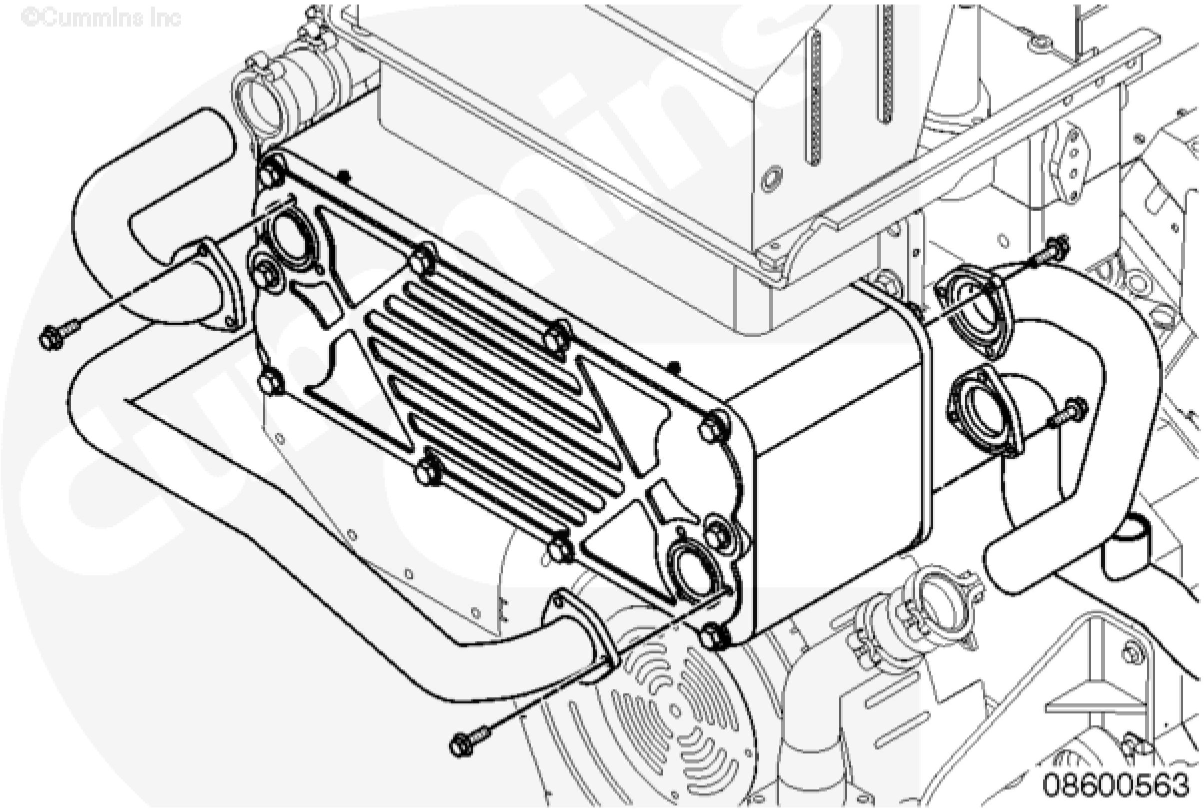
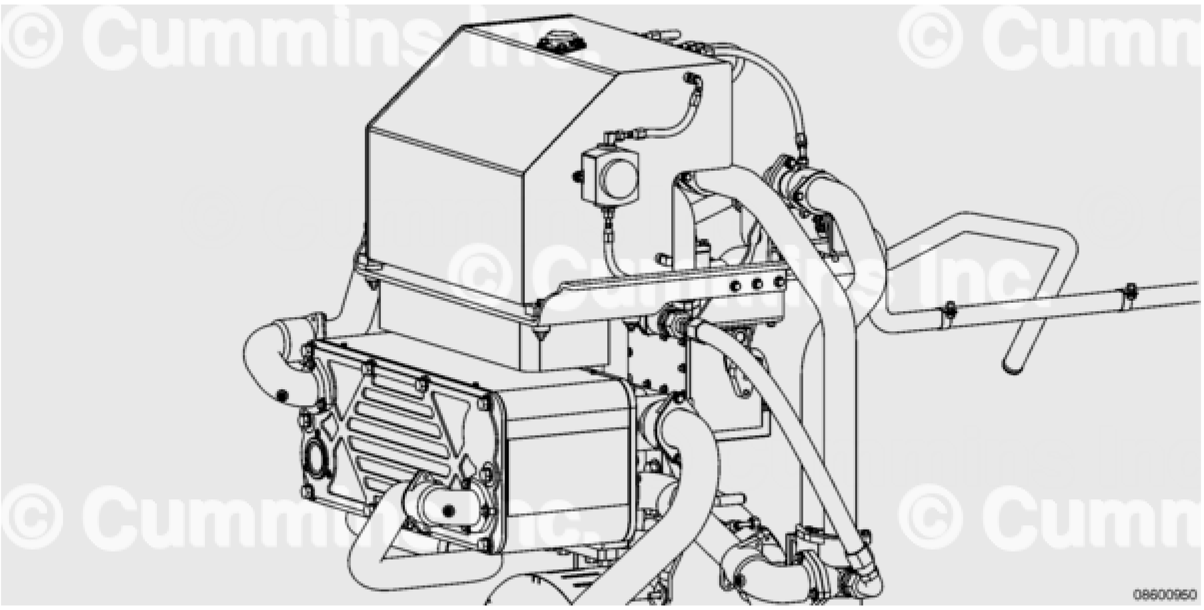


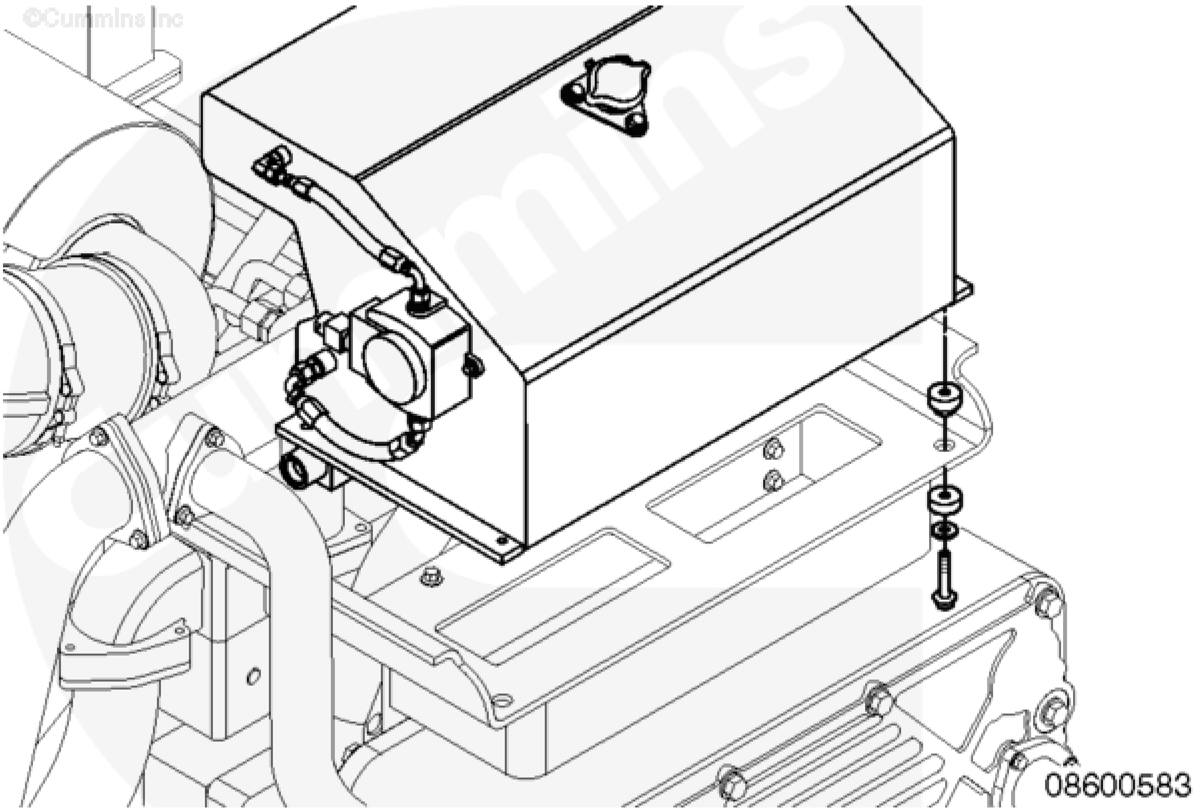
Plate Type Heat Exchanger

The heat exchanger used on the QSK60 Marine engine with electronically actuated injectors is also a titanium plate design that mounts on the front of the engine. It is similar in design and layout as the heat exchanger used on the mechanically actuated injector engine, but the cooling connections have been improved to utilize a casting and captured-tube design, which improves the reliability of the plumbing of the heat exchanger. The design features a casting which is assembled to the heat exchanger, which mates to a separate tube using an o-ring and wear sleeve interface to both isolate and seal the connection point.



QSK60 with Electronically Actuated Injectors Heat Exchanger Coolant Connections

The expansion tank is **not** integrated with the heat exchanger assembly. It is mounted above the heat exchanger. The expansion tank does **not** need to be removed to disassemble the plate pack, but **must** be removed in order to remove the entire heat exchanger assembly.

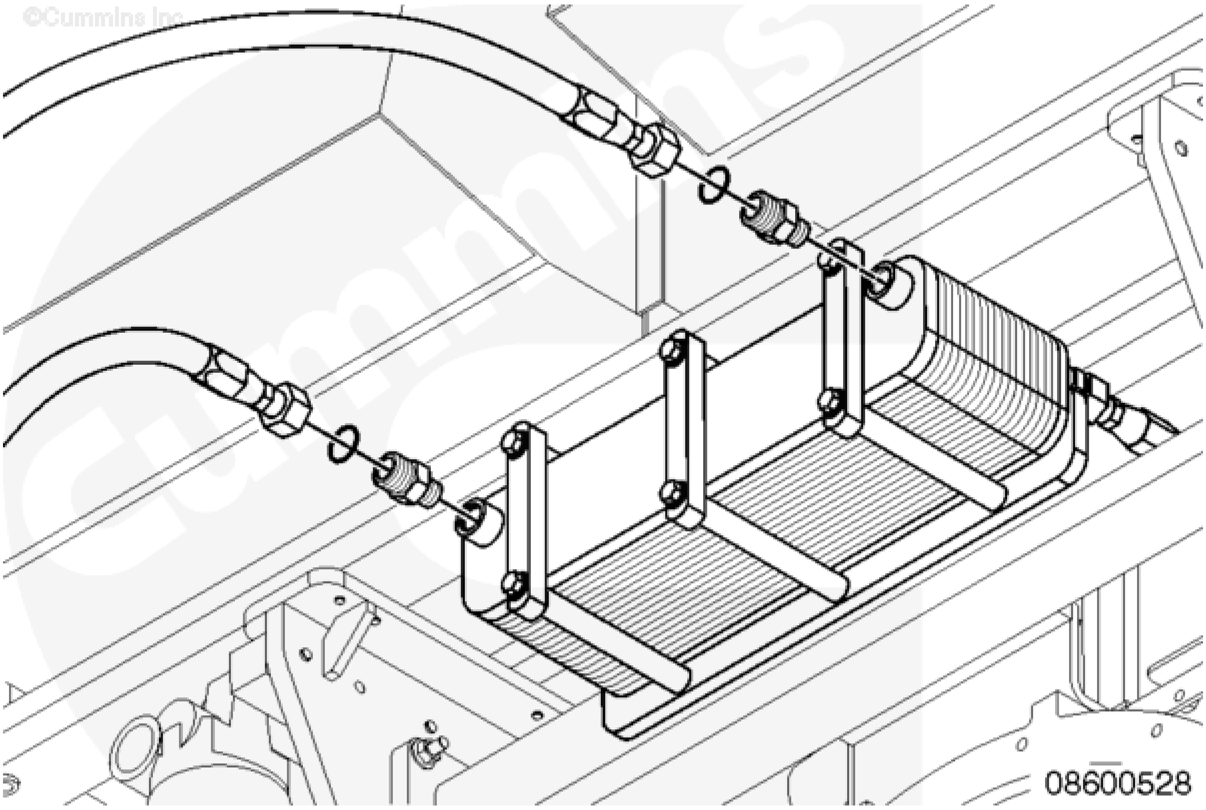


Expansion Tank

The keel-cooled engines have flexible connections that connect to the ship board piping. There are two separate keel coolers. The LTA pump for the keel-cooled engines is driven from the front of the hydraulic gear drive.

For auxiliary installations that require radiator cooling, the engine comes with a mounted fan drive. The fan and radiator **must** be customer-supplied.

The marine gear oil cooler is an option on the propulsion engine and is mounted on the block above the flywheel housing and below the air cleaner assembly. The cooler is part of the LTA circuit. The marine gear oil connections to the cooler are customer-supplied.



Marine Gear Oil Cooler

Last Modified: 18-Nov-2008